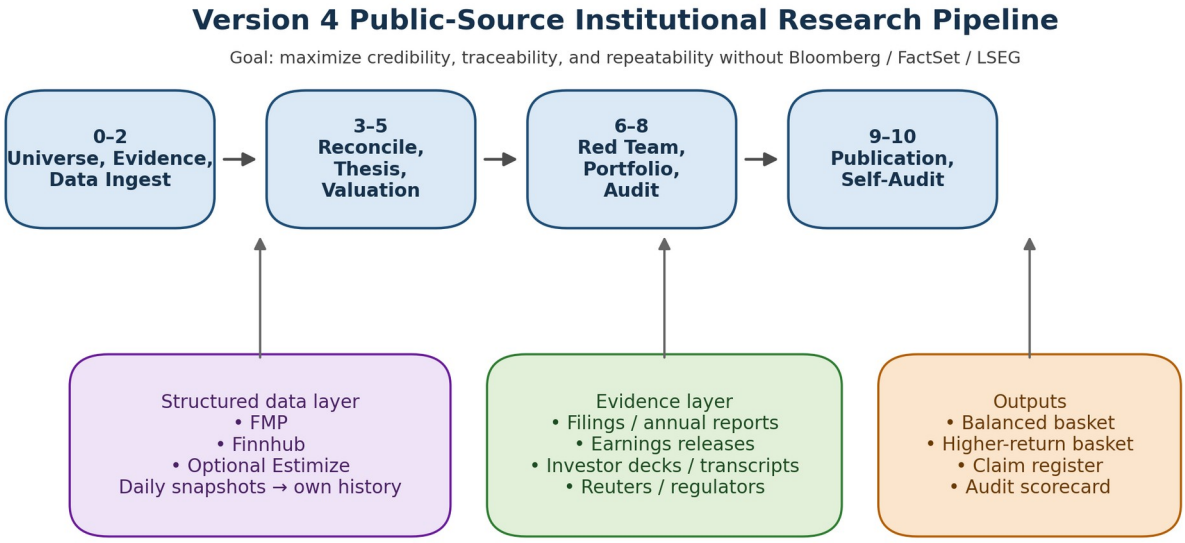
Version 4 Research Pipeline Architecture & Operating Manual

Public-source institutional-style workflow for AI / energy / infrastructure investing
Designed to maximize credibility without Bloomberg / FactSet / LSEG

What this document does

- Defines the operating architecture, source hierarchy, data stack, and governance model.
- Specifies the two API-fed datasets to add: FMP and Finnhub, with optional Estimize overlay.
- Explains what the process can and cannot claim, and where paid institutional datasets would still add value.



Executive summary

Scope

Version 4 is a claim-led research pipeline built to solve the weaknesses identified in earlier drafts: overconfident prose, inconsistent source quality, weak valuation discipline, and no hard publication gate. The upgraded design separates evidence gathering, structured data ingestion, analysis, red-team challenge, portfolio construction, and final publication so that each stage has a narrow mandate and auditable outputs.

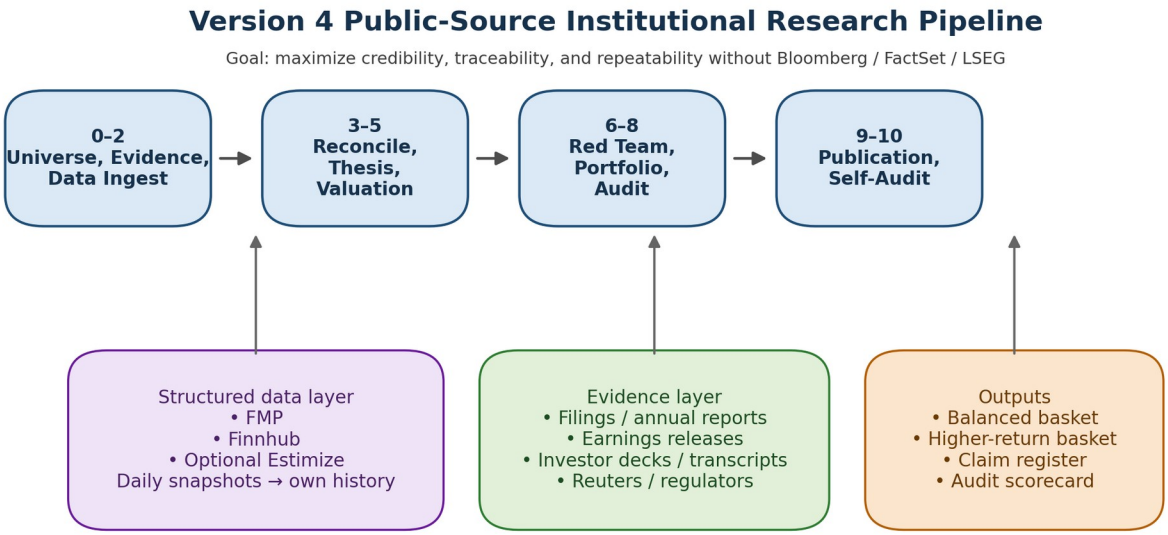
- **Primary objective** — produce friend-facing or client-ready thematic investment reports that are defensible, traceable, and repeatable.
- **Core data stack** — Tier 1 company disclosures + Tier 2 independent verification + Tier 3 API-fed market/consensus data.
- **Two API datasets** — FMP for estimates, price targets, ratings snapshots, ratios, and quotes; Finnhub for recommendation trends, price targets, EPS surprises, and event metadata.
- **Hard rule** — no final document can publish if any core claim lacks a source class, date, or audit status.

Layer	What it provides	Why it matters	Publish rule
Tier 1: Primary facts	Filings, annual reports, earnings releases, decks, transcripts, exchange notices	Anchors factual claims in issuer-controlled documents	Required for core company facts
Tier 2: Independent verification	Reuters, regulators, system/operator data, government sources	Checks whether important claims hold up outside management framing	Required for major catalysts and contracts where available
Tier 3: Structured data APIs	FMP and Finnhub; optional Estimize later	Adds machine-readable prices, estimates, ratings and history snapshots	Allowed for consensus, targets, quotes, revisions proxies
Tier 4: House outputs	Scenario ranges, rankings, weights, expected return bands	Makes judgment explicit instead of disguised as fact	Must always be labelled as house view

System architecture

Design

The pipeline is intentionally modular. Evidence is assembled first, quantitative data is ingested second, and only then are investment cases written. This sequencing matters: it stops polished writing from outrunning the source base.



Module	Owner / role	Primary artefacts
Evidence Librarian	Source-control stage	Claim ledger, source inventory, corroboration flags
Structured Data Ingest	Data stage	Daily snapshots, consensus tables, ratings history
Analyst + Valuation engines	Investment synthesis	Stock notes, valuation scores, return bands
Red Team + Auditor	Control stage	Disconfirming memos, pass/fail audit record
Publication Synthesiser	Output stage	Readable final note, appendices, scorecard

Data stack	Feeds

Version 4 adds two machine-readable datasets so the process can maintain its own running history rather than relying on one-off manual lookups.

Dataset	Use in pipeline	Core endpoints / data families	Assessment
FMP	Base quantitative layer	Analyst estimates, price-target consensus, ratings snapshots, ratios, quotes	Best first API addition: broad utility and easy to store historically
Finnhub	Cross-check and event layer	Recommendation trends, price targets, EPS surprises, earnings calendar, quote support	Adds independent comparison and revision-style signals
Estimize (optional)	Alternative-consensus overlay	Crowdsourced revenue / EPS expectations via API	Helpful later, but not required for Version 4 launch

Koyfin remains useful as a human-facing workstation, but because it does not provide API access and restricts download of estimates and valuation data for global equities, it is not part of the automated spine of this process.

Operating cadence

Rhythm

Frequency	What runs	Output
Daily	API ingest from FMP and Finnhub; quote refresh; estimate / target snapshot; event calendar update	New snapshot tables and change log
Weekly	Analyst refresh; valuation crowding review; red-team watchlist update	Re-ranked ideas and watchlist memo
On report day	Claim re-verification; full audit; publication synthesis; self-score	Final report only if all gates pass

Publication gates

Control

- **Fail conditions** — undefined audit fields, missing dates, retail-source-only core claims, management guidance presented as fact, or forecast ranges without methodology tags.

- **Pass with disclosure** — a point is directionally supported but lacks ideal independent confirmation; the note can publish only if that limitation is disclosed in-line or in the appendix.
- **Full pass** — all material claims are mapped to the claim ledger, data snapshots reconcile within tolerance, and the self-audit score clears the target threshold.

Gate	Threshold	Action if breached
Source strength	Every material claim tagged with Tier, date, and evidence class	Return to Evidence Librarian
Data reconciliation	FMP vs Finnhub mismatch within tolerance or explicitly disclosed	Return to Data Reconciliation Engine
Valuation transparency	Every expected-return band linked to method and scenario labels	Return to Valuation Engine
Audit completeness	Zero undefined / blank control fields	Block publication

What this process can credibly claim

Boundaries

- **Can claim** — strong public-source factual reliability, consistent source provenance, explicit judgment tags, cleaner expected-return framing, and repeatable portfolio construction.
- **Cannot claim** — true Bloomberg / FactSet / LSEG consensus parity, full historical broker revision history, or personalised financial advice tailored to an individual's tax and risk profile.
- **Why this still matters** — the largest improvement comes from process discipline, not from pretending to own premium datasets that the workflow does not actually have.

Target scorecard

Outcome

Dimension	Current legacy process	Version 4 target	What drives the improvement
Process quality	~6.6/10	8.7/10	Hard gates, modular stages, clearer responsibilities
Evidence quality	~7.2/10	8.6/10	Claim ledger, source hierarchy, independent verification

Dimension	Current legacy process	Version 4 target	What drives the improvement
Investment usefulness	~6.8/10	8.4/10	Valuation stage, red team, portfolio constraints, API-fed snapshots
Institutional credibility	~6.0/10	8.3/10	Less overclaiming, stronger disclosure, cleaner data lineage

Recommended implementation order

- Build the claim ledger and publication gate first.
- Add FMP and Finnhub ingestion with daily snapshot storage.
- Stand up valuation, red-team, and reconciliation stages.
- Only then upgrade the final report packaging and visuals.
- Add Estimize later if you want alternative-consensus signal rather than core pipeline functionality.

Bottom line: Version 4 is the strongest credible non-terminal process for this use case. It will not equal Bloomberg, but it can produce research that is significantly more defensible, more auditable, and more useful than the current workflow.